Assignment 1: Kicking

Due date: Tuesday, 03.11.2026 at 23:59

1 Introduction

In this assignment, you will implement a robot agent that can autonomously shoot goals in the rSoccer simulator. The objective is to design a policy that allows the robot to score as many goals as possible within a limited time window.

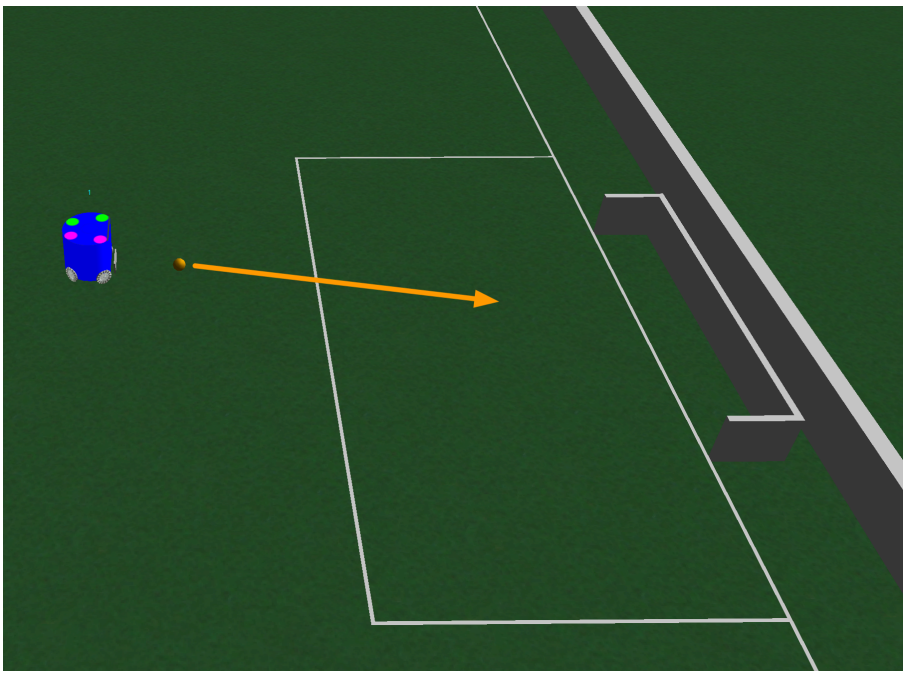


Figure 1: Robot shooting at the goal (image taken from grSim simulator)

2 Task

Your goal is to develop an agent that can shoot the ball into the opponent's goal as efficiently as possible. Once a goal is scored or the ball has left the field, the ball will be repositioned randomly on the field, and the robot must quickly adapt and prepare for the next shot.

2.1 Learning to Shoot

Your robot should learn a shooting policy that decides:

- How to approach the ball.
- How to align itself and prepare the shot.
- When and how to shoot the ball toward the goal.

Note: The kicker will only have an effect on the ball when the kicker touches the ball. rSoccer provides an infrared sensor which tells you when the ball touches the kicker.

2.2 Score goals against an active goalkeeper

Start by testing your striker in the easier setup, where the goal is undefended. This helps you verify that your policy can reliably approach the ball, align, and shoot accurately.

After that, evaluate your striker against the provided goalkeeper strategy. This gives a more realistic challenge and helps you improve robustness under defensive pressure.

Use the following runner:

- `run_rsoccer_exercise1.py`

You can switch the opponent goalkeeper implementation in the runner by selecting the yellow strategy (for example, idle goalkeeper first, then the provided rule-based or RL-based goalkeeper - see README from API repository).

Technical Approach

You can choose any technical approach to complete the assignment. However, learning-based methods will be rated more highly than simple heuristics.

Evaluation

Your agent will be evaluated based on how many goals it can score in a 90-second simulation run. First without the goalkeeper and then with a goalkeeper. After each goal, the ball will automatically spawn at a new random position on the right side of the field.

3 Submission Details

Submit your code via Ilias until the due date. Late submissions will not be accepted. Upload your presentation file on Ilias before the start of the lab session.

In the next lab session on Wednesday (04.11.2026) your team will:

- Give a short presentation (maximum 10 minutes) explaining the technical implementation of your shooting policy.
- Answer questions about your approach from the supervisors and other teams.
- Your strategy will be tested in a live 90-second demo where your striker attempts to score as many goals as possible both without a goalie and with a goalie.

4 Grading

- **Technical Approach (50%):** Quality, originality, and suitability of the implemented shooting skill. Approaches that go beyond minimal or off-the-shelf solutions and show thoughtful design will be rated more highly.

- **Scenario Performance (20%):** Number of goals scored within the given time frame in the live demonstration.
- **Presentation (30%):** Clarity of explanation, ability to answer questions, and insights into design decisions.